

Samuel Liu

Senior Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 9 years of experience in developing scalable, high-performance backend systems. Proficient in **JavaScript**, **TypeScript**, **Java** and **Kotlin** with a strong focus on API design, server performance optimization, and cloud-based microservices architectures. Skilled in **DevOps** practices including CI/CD, containerization, and cloud deployment with **AWS**, **Docker**, and **Kubernetes**. Adept at working with SQL (**PostgreSQL**, **MySQL**) and NoSQL (**MongoDB**, **DynamoDB**) databases. Recognized for delivering clean, maintainable code and solving complex technical challenges in fast-paced environments.

SKILLS

- Languages: Java, TypeScript, Python, Kotlin
- Frontend: React, React-Native, Next.js, Electron.js, TanStack, MUI, Tailwind CSS, Shadcn
- Backend: Node.js, ExpressJS, NestJS, Spring Boot, GraphQL
- AI & ML: Generative AI, OpenAI API, LLMs, Diffusion Models, RAG, Pinecone, LangChain
- Databases: MySQL, PostgreSQL, MongoDB
- DevOps: Docker, Kubernetes, AWS, Cloudflare, DigitalOcean
- Testing: Jest, Playwright, Cypress
- Tools: Storybook, PostHog, Sentry, Figma

PROFESSIONAL EXPERIENCE

Senior Full Stack Engineer

11/2022 - present

Azumo

Built an AI-powered creative platform with **React**, **TypeScript** and **Node.js**.

- **Built advanced interactive canvas features** (zooming, panning, node selection, hotkeys) powered by **React Flow**, optimizing state management and performance to support **450+ real-time users** and **1,000 concurrent nodes**.
- **Migrated and refactored 120+ React components** from JSX to **TypeScript (TSX)**, enhancing scalability, maintainability, and type safety across the frontend codebase.
- **Implemented collaboration and versioning systems** including real-time canvas sharing, cross-canvas copy/paste, and version tracking to boost team productivity and user retention.
- **Engineered core infrastructure and deployment pipelines** using **Node.js**, **PostHog**, **DigitalOcean**, and **feature flagging**, cutting production issue resolution times by **40%**.
- **Delivered exceptional UX and analytics** through **TailwindCSS theming**, **D3.js dashboards**, and **Shepherd.js onboarding tours**, increasing onboarding success by **60%** and reducing user support queries by **30%**.
- **Engineered extensive unit and integration test suites** leveraging **Jest** and **Cypress**, achieving an exceptional 90% test coverage to ensure optimal application reliability, performance, and maintainability.

Senior Software Engineer

04/2021 – 09/2022

Sutter Health

- **Implemented a doctor-patient chat feature** in the Android app (Kotlin), leveraging WebSocket connections for instant two-way communication.
- **Added rich messaging functions** like read receipts and push notifications to enhance user engagement, which end-to-end encryption ensuring the confidentiality of sensitive health discussions.
- **Utilized WebRTC's built-in end-to-end encryption** (DTLS and SRTP) and provided options for both video and audio-only calls to accommodate user preferences and bandwidth constraints.
- **Provided constructive feedback** on others' pull requests through code reviews, enhancing overall code quality and collaboration.

Software Engineer

05/2018 - 03/2021

Sutter Health

- **Designed and implemented RESTful API endpoints** in Java to power the messaging and telehealth features.
- **Enhanced telehealth features** by adding WebRTC-based video and audio communication for remote consultations.
- **Refactored legacy REST endpoints into GraphQL queries and mutations**, enabling more efficient data retrieval and better support for mobile clients.
- Collaborated closely with front-end and mobile teams to define **GraphQL queries and fragments**, improving feature delivery velocity and reducing back-and-forth.
- **Containerized and orchestrated services** with Docker and Kubernetes, reducing deployment times and facilitating CI/CD with GitHub Actions.
- **Ensured HIPAA compliance** and secure data handling through secure data pipelines and adherence to industry standards.
- **Collaborated in Agile teams** to translate requirements, improve team productivity, and provide mentoring for junior engineers on best practices.

Software Engineer

06/2013 - 04/2018

Amazon

- **Implemented scalable backend microservices** in Java (**Spring Boot**) and **AWS Lambda**, powering Prime Video's **home feed**, **search results**, and **"Continue Watching"** modules, serving millions of daily active users globally.
- **Optimized video encoding workflows** by integrating **FFmpeg**, **AWS Elastic Transcoder**, and later **MediaConvert**, reducing encoding times by 30% while maintaining consistent adaptive bitrate quality.
- **Implemented content delivery optimizations** with **AWS CloudFront**, **Route 53**, and custom caching strategies to reduce global streaming latency and buffer rates.
- **Contributed to the Prime Video web frontend** by developing **React components** for the **Home**, **Details**, and **Playback** pages, ensuring consistent cross-platform rendering between browsers and smart TVs.
- **Worked closely with UX and design teams** to implement **accessibility (WCAG 2.0)** and internationalization (i18n) features for subtitles, localization, and right-to-left (RTL) support.

Software Development Engineer Intern

05/2012 - 08/2012

Amazon

- **Collaborated with senior engineers** to refactor a legacy internal module, improving performance and cutting average response times by 40%.
- **Built RESTful APIs for an internal platform**, enabling seamless data exchange between microservices and downstream analytics systems.
- **Integrated monitoring and logging** using **AWS CloudWatch** and internal metrics libraries, enhancing visibility into production health.

EDUCATION

Bachelor's Degree in Computer Science

08/2009 - 05/2013 | Champaign, IL

University of Illinois Urbana-Champaign